Snake Game Deluxe Web Edition

B.Tech CSE (2025–2029)
Semester II – Web Development
Technologies Used:
HTML5 • CSS3 • JavaScript

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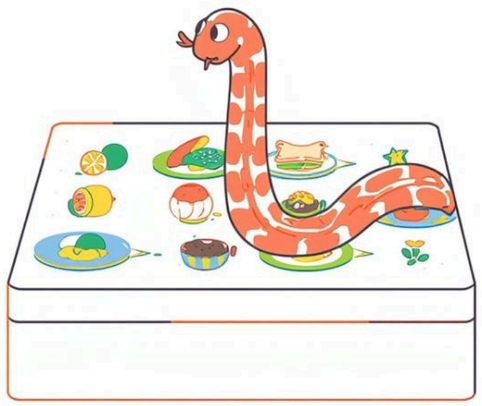
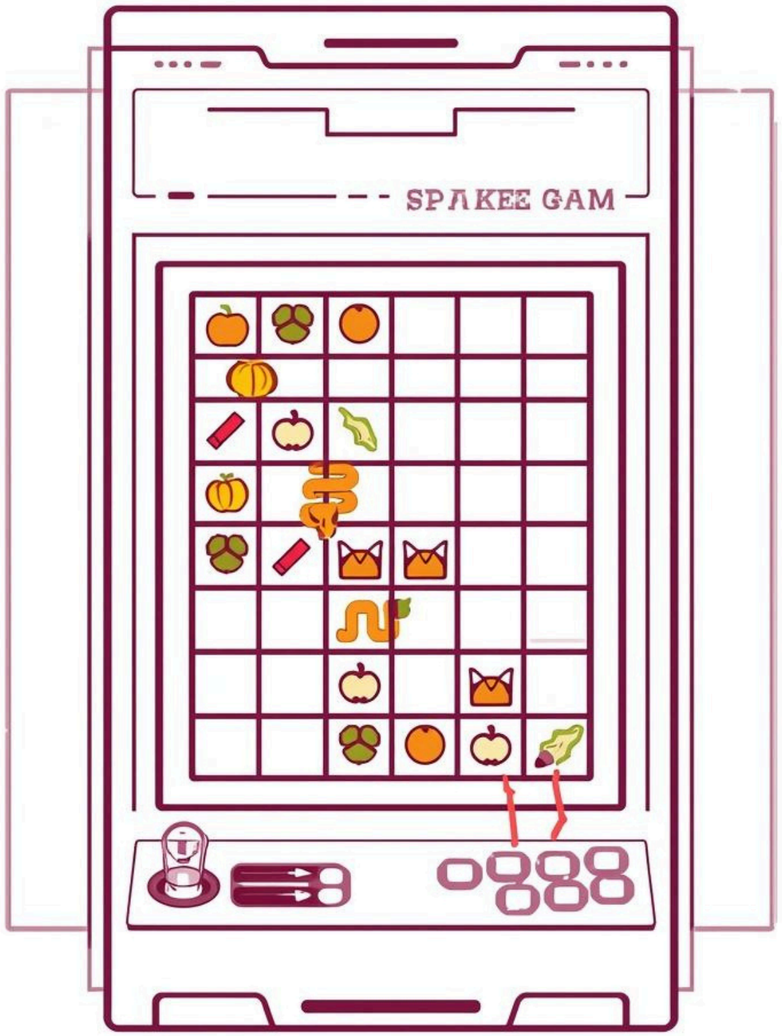


Index Overview

- Problem Statement: Modernize the classic Snake game experience
- Your Solution: Showcase key features and technologies used
- Working: Explain game flow and technical approach outlined



Classic snake games available online are often very basic and lack modern features.

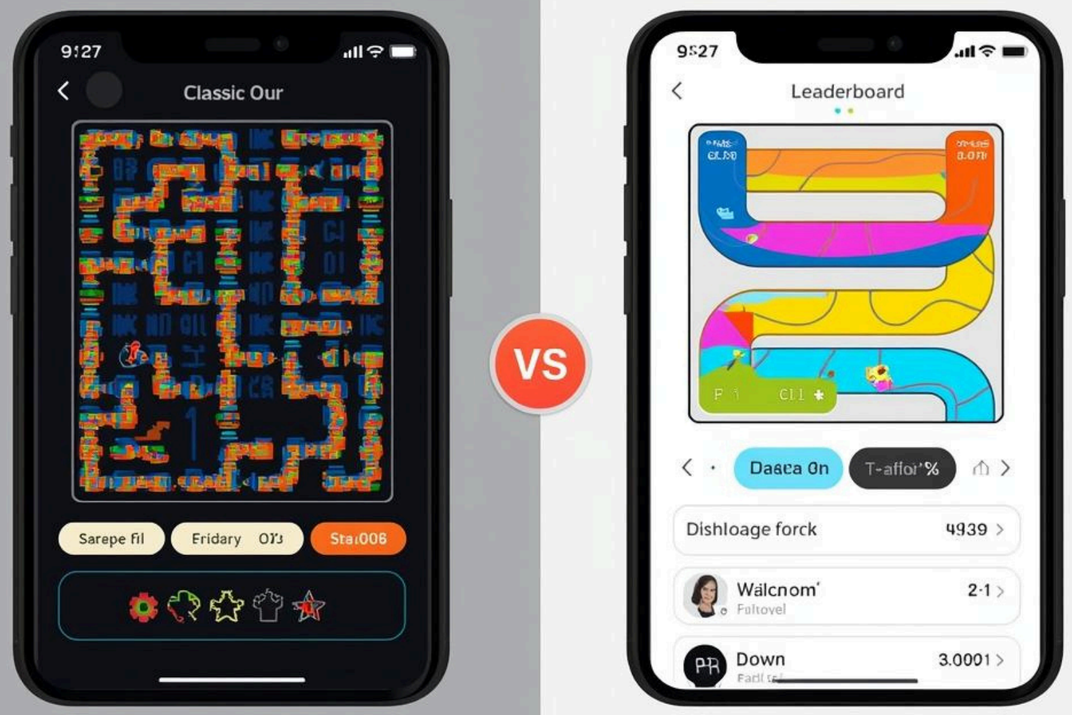
Most versions do not include:

- Real-time score tracking
- Leaderboards to compare player scores
- Community interaction between players
- Modern user interface design

Because of these limitations, players cannot compete with others or share strategies.

Objective:

Develop a responsive Snake Game web application that includes score tracking, leaderboard functionality, and a forum section for user interaction.



To solve this problem, a Snake Game Deluxe Web Application was developed using HTML, CSS, and JavaScript.

Key Features:

- Canvas-based game board
- Snake movement using keyboard arrow keys
- Random food generation
- Score tracking system
- Increasing game speed with higher scores
- Obstacle mode for higher difficulty
- Leaderboard using browser localStorage
- Forum section where players can post comments

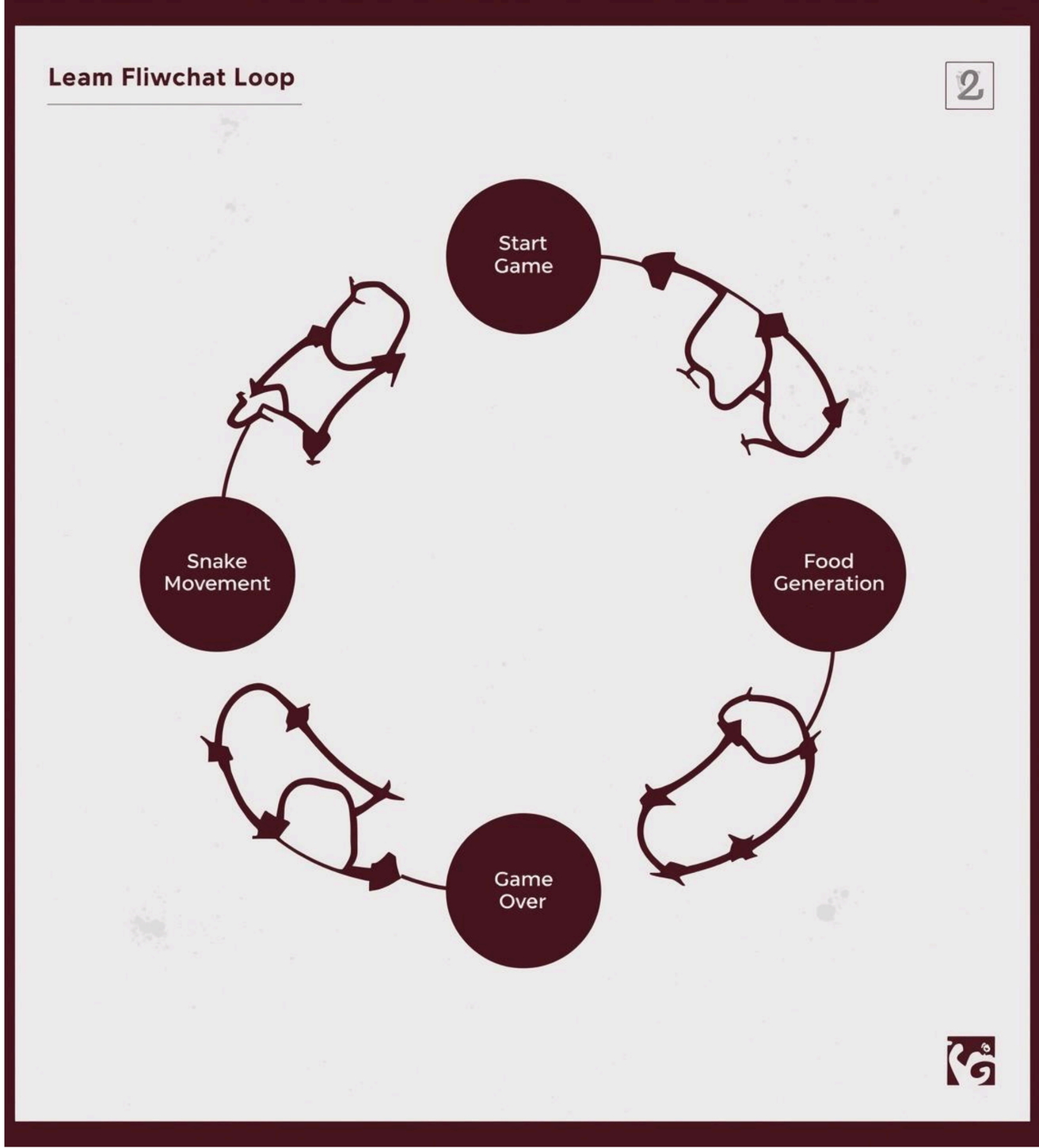
The application runs entirely in the browser without backend integration.

The game operates using a continuous game loop implemented in JavaScript.

Game Flow:

Start Game
↓
Snake moves using arrow keys
↓
Food appears randomly on the canvas
↓
Snake eats food and score increases
↓
Snake grows longer
↓
Game speed increases as score rises
↓
Collision detection checks wall, body and obstacles
↓
Game Over screen appears
↓
Player score saved in leaderboard

The game is rendered using the HTML Canvas element.

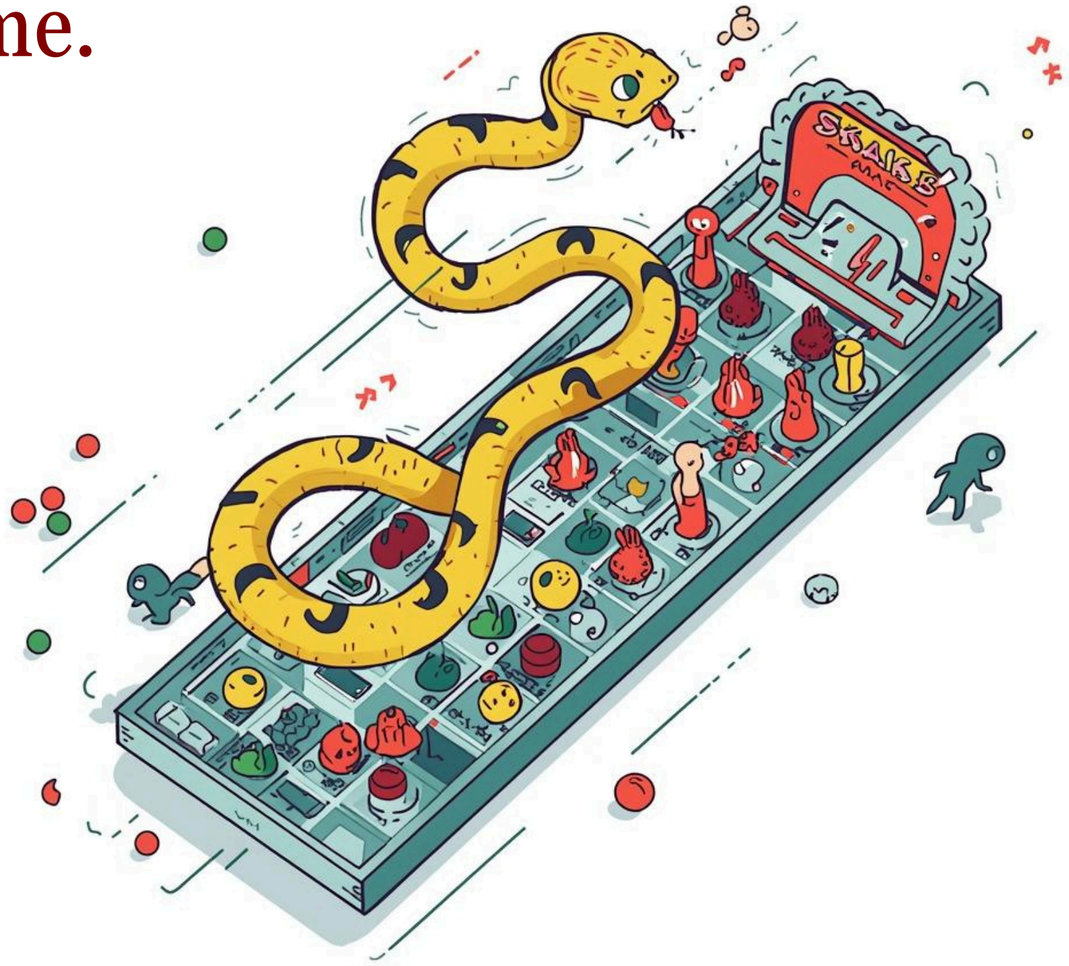


The Snake Game Deluxe Web Edition successfully demonstrates the development of an interactive browser-based game.

The project improves the traditional snake game by introducing features such as:

- Score tracking
- Leaderboard comparison
- Obstacle gameplay
- Forum interaction

This project demonstrates the use of HTML, CSS and JavaScript to build an engaging web-based arcade game.
It also highlights important programming concepts such as game loops, collision detection, and browser data storage using localStorage.



Thank You

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